# Forcing-sterility of the realizable unit-distance lineage and a codegree obstruction to $\chi \geq 6$

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**Abstract**

The chromatic number of the plane, $\chi(\mathbb{R}^2)$, is known only to lie in $[5, 7]$. The lower bound $\chi(\mathbb{R}^2) \geq 5$ rests on finite unit-distance graphs (UDGs): de Grey (2018) exhibited the first such graph with $\chi \geq 5$, and the Polymath16 project reduced the order to roughly 510 vertices. A natural route to $\chi(\mathbb{R}^2) \geq 6$ would locate, somewhere in this realizable lineage of $\chi = 5$ UDGs, a non-adjacent vertex pair forced to distinct colors in every proper 5-coloring (a *clamp*); realizing such a pair lifts the chromatic number to 6. This note records a negative structural result: the realizable lineage contains no such forcing, and provably so. We recall an elementary lemma (the *Essential-Pair Lemma*, essentially known) which implies that a vertex-critical 5-chromatic graph cannot host any forced non-adjacent pair. We confirm the consequence by an exhaustive computational census: all 1,955,948 non-adjacent pairs across the nine $\chi = 5$ lineage graphs are free, with zero forced pairs of either kind. Since the lineage is vertex-critical, this is a one-line corollary of the Lemma, and the deeper point is conceptual: the lineage was produced by SAT-driven minimization, which drives graphs toward vertex-criticality, and criticality deletes exactly the vertices that could host forcing. The lineage is forcing-sterile *by construction*. We then frame the search for the missing $\chi \geq 6$ object through a codegree obstruction: every planar UDG is $K_{2,3}$-free, so its edge count is bounded by $m \leq n\big(1 + \sqrt{8n-7}\big)/4$; combined with the Kostochka-Yancey lower bound for 6-critical graphs and the vertex Folkman number $F_v(2, 2, 2, 2, 2; 4) = 16$, this excludes every small dense Folkman-type host. The missing object must lie in the class of graphs that are simultaneously $K_4$-free and $K_{2,3}$-free, of order at least roughly 26. We claim no new bound on $\chi(\mathbb{R}^2)$.

## 1 Introduction

The Hadwiger-Nelson problem asks for the chromatic number $\chi(\mathbb{R}^2)$ of the *unit-distance graph* on the Euclidean plane: vertices are the points of $\mathbb{R}^2$, and two points are adjacent iff they are at distance exactly 1. The classical bounds are

$$5 \leq \chi(\mathbb{R}^2) \leq 7,$$

and have resisted improvement for three quarters of a century. The upper bound $\chi(\mathbb{R}^2) \leq 7$ is the Isbell hexagonal coloring (tile the plane with regular hexagons of diameter slightly less than 1 and color in a repeating 7-pattern). The lower bound $\chi(\mathbb{R}^2) \geq 5$ is recent and computational: de Grey [1] constructed a finite unit-distance graph on 1581 vertices with $\chi \geq 5$, verified by a SAT proof, and the Polymath16 project [5] subsequently reduced the order of such graphs to roughly

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510 vertices. We refer to this family of explicitly realized $\chi = 5$ UDGs as the *realizable lineage*. For background on the problem and its history see Soifer [6].

The natural finitary route to $\chi(\mathbb{R}^2) \geq 6$ is to find a finite UDG with $\chi \geq 6$. One concrete mechanism for manufacturing such a graph out of a $\chi = 5$ graph is a *clamp*.

**Definition 1** (Clamp)**.** Let $G$ be a 5-chromatic graph and $(s, t)$ a non-adjacent pair of vertices. We call $(s, t)$ a *clamp* if every proper 5-coloring of $G$ assigns $c(s) \neq c(t)$.

If a clamp $(s, t)$ in a UDG can be *realized*, that is, if the geometry permits placing a vertex at unit distance from both $s$ and $t$ while preserving the unit-distance structure (so that an edge effectively appears across the forced-different pair), the chromatic number is lifted to 6. The motivating question for this note is therefore:

> *Does the realizable lineage of $\chi = 5$ UDGs contain a clamp (or any other forced non-adjacent pair) that could be exploited to reach $\chi \geq 6$?*

The answer of this note is **no, and provably so**. We make this precise through one elementary lemma (Section 3), confirm it with an exhaustive computational census (Section 4), explain it structurally by way of vertex-criticality (Section 5), and then reframe the search for the genuinely missing object through a codegree obstruction (Section 6). Section 7 synthesizes the picture and states the target object honestly.

**Scope and honesty.** This is a negative, explanatory note. We move no bound on $\chi(\mathbb{R}^2)$. The contribution is threefold: (a) the observation that the realizable lineage is forcing-sterile *by construction*, obtained by applying a known elementary lemma to its vertex-criticality; (b) an exhaustive computational census confirming it across the entire lineage; and (c) a codegree framing that locates the missing $\chi \geq 6$ object outside the reach of both the realizable lineage and the small dense graphs one would naively manufacture. None of the three ingredients is individually deep; the synthesis is the point.

## 2 Preliminaries

All graphs are finite, simple, and undirected. For a graph $G$ we write $V(G)$ for its vertex set, $n = |V(G)|$, $m = |E(G)|$, and $\chi(G)$ for the chromatic number. A graph is 5-*chromatic* if $\chi(G) = 5$. A proper $k$-coloring is a map $c : V(G) \to \{1, \ldots, k\}$ with $c(u) \neq c(v)$ whenever $uv \in E(G)$.

**Definition 2** (Vertex-critical)**.** A graph $G$ is *vertex-critical* (with respect to its chromatic number) if $\chi(G - v) < \chi(G)$ for every $v \in V(G)$. A 5-chromatic graph is vertex-critical iff $\chi(G - v) \leq 4$ for every vertex $v$.

**Definition 3** (Forcing of a non-adjacent pair)**.** Let $G$ be a 5-chromatic graph and let $(s, t)$ be a pair of distinct, *non-adjacent* vertices. We classify $(s, t)$ as follows.

- Forced-same: every proper 5-coloring of $G$ has $c(s) = c(t)$.
- Forced-different: every proper 5-coloring of $G$ has $c(s) \neq c(t)$.
- Forced: either forced-same or forced-different holds.
- Free: neither holds, i.e. there exist a proper 5-coloring with $c(s) = c(t)$ and a (possibly different) proper 5-coloring with $c(s) \neq c(t)$.

A clamp (Definition 1) is exactly a forced-different non-adjacent pair. A forced-same pair is no weaker as a route to $\chi \geq 6$: adding a single unit edge across a forced-same pair produces a forced-different pattern elsewhere (a splice argument), so forced-same and forced-different are interconvertible as $\chi \geq 6$ ingredients. For the purposes of this note the relevant dichotomy is simply *forced* versus *free*: a single forced pair anywhere in the lineage would be a candidate $\chi \geq 6$ ingredient.

**Deciding forcing by SAT.** Each forcing question reduces to two $k$-colorability decisions, both decidable by a standard $k$-coloring SAT encoding.

- FORCED-DIFFERENT holds for $(s, t)$ iff the graph $G/\{s, t\}$, obtained by *identifying* (merging) $s$ and $t$, is not 5-colorable. Merging forces $c(s) = c(t)$; if even that constrained instance is 5-UNSAT, then no proper 5-coloring has $c(s) = c(t)$, so every proper 5-coloring has $c(s) \neq c(t)$.
- FORCED-SAME holds for $(s, t)$ iff the graph $G + st$, obtained by *adding* the edge $st$, is not 5-colorable. Adding the edge forbids $c(s) = c(t)$; if that instance is 5-UNSAT, then no proper 5-coloring has $c(s) \neq c(t)$, so every proper 5-coloring has $c(s) = c(t)$.
- Otherwise $(s, t)$ is FREE.

Both tests are decisive: a 5-UNSAT verdict from a complete solver is a proof, and a satisfying assignment is an explicit certificate of freeness on that side.

**Control objects.** Throughout we keep in mind the structural controls of the program. The unit-distance graph on $\mathbb{Q}^2$ has chromatic number 2 (Woodall [8]); any purely combinatorial argument for $\chi(\mathbb{R}^2) \geq 5$ must use the topology of $\mathbb{R}$ in a way that fails on $\mathbb{Q}^2$. The codegree obstruction of Section 6 is genuinely Euclidean: it uses two-circle rigidity, which fails under the sup-norm (where unit "circles" are squares and can share a segment). The Essential-Pair Lemma of Section 3, by contrast, is pure graph theory and claims no bound anywhere.

## 3 The Essential-Pair Lemma

The following lemma is elementary and essentially known. Its corollary, that a vertex-critical $k$-chromatic graph admits no forced non-adjacent pair, appears as Theorem 3.17 ("there are no implicit relations in critical graphs") of Martin [4], where forced-different and forced-same pairs are termed *implicit-edges* and *implicit-identities*; the single-deletion form we state is a one-line sharpening of his Theorems 3.2 and 3.17. The result is also implicit in standard critical-graph theory (Dirac and Gallai; see Jensen and Toft [2]). We state and prove it for self-containedness and make no priority claim; the proof uses only the definition of a proper coloring.

**Lemma 1** (Essential-Pair Lemma)**.** *Let $G$ be a 5-chromatic graph and let $(s, t)$ be a non-adjacent pair that is* forced *(forced-same or forced-different). Then*

$$\chi(G - s) \geq 5 \quad \textit{and} \quad \chi(G - t) \geq 5.$$

*Proof.* We prove $\chi(G - s) \geq 5$; the statement for $t$ follows by symmetry.

Suppose toward a contradiction that $\chi(G - s) \leq 4$. Fix a proper 4-coloring of $G - s$ using colors $\{1, 2, 3, 4\}$. We extend it to $G$ in two different ways.

*The "different" pattern.* Give $s$ the new color 5. Every neighbor of $s$ lies in $G - s$ and so is colored with a color in $\{1, 2, 3, 4\}$; hence no neighbor of $s$ has color 5, and the extension is a proper

5-coloring of $G$. In it, $c(s) = 5$. Since $t \neq s$ and $t$ is colored with a color in $\{1, 2, 3, 4\}$, we have $c(s) \neq c(t)$. Thus the forced-different pattern is realized.

*The "same" pattern.* Starting again from the same proper 4-coloring of $G - s$, set $c(s) = 5$ as before, and additionally *recolor* $t$ to 5. Because $s$ and $t$ are **non-adjacent**, every neighbor of $t$ also lies in $G - s$ (a neighbor of $t$ cannot be $s$), and so is colored in $\{1, 2, 3, 4\}$; recoloring $t$ to 5 therefore creates no monochromatic edge at $t$. The vertex $s$ likewise has all neighbors in $\{1, 2, 3, 4\}$, so no edge at $s$ is monochromatic either. The result is a proper 5-coloring of $G$ with $c(s) = 5 = c(t)$. Thus the forced-same pattern is realized.

Both the same-colored and the differently-colored patterns occur in proper 5-colorings of $G$. Hence $(s, t)$ is FREE (Definition 3), contradicting the hypothesis that it is forced. Therefore $\chi(G - s) \geq 5$, and by symmetry $\chi(G - t) \geq 5$. □

**Corollary 1.** *A vertex-critical* 5*-chromatic graph contains* no *forced non-adjacent pair: every non-adjacent pair is free.*

*Proof.* Let $G$ be vertex-critical and 5-chromatic, so $\chi(G - v) \leq 4$ for every vertex $v$. If some non-adjacent pair $(s, t)$ were forced, then Lemma 1 would give $\chi(G - s) \geq 5$, contradicting $\chi(G - s) \leq 4$. Hence no non-adjacent pair is forced. □

The content of Corollary 1 is intuitive once stated: a forced pair $(s, t)$ requires that the rest of the graph be "heavy" enough at both $s$ and $t$ to constrain their colors even without an edge between them, and that heaviness is precisely what survives the deletion of $s$ (or $t$). If the graph is already minimal in the sense that every vertex deletion drops the chromatic number, no such residual heaviness exists.

## 4 The exhaustive census

We now report the computational confirmation. The realizable lineage of $\chi = 5$ UDGs studied here consists of nine graphs, of orders

$$510,\ 517,\ 529,\ 553,\ 610,\ 633,\ 803,\ 826,\ 874,$$

descending from the de Grey / Polymath16 constructions. (An earlier triage table listed three further graphs, on 199, 403, and 721 vertices, which on re-examination have $\chi = 4$; they are excluded here, and their exclusion does not affect the census headline.)

**Method.** For every non-adjacent pair $(s, t)$ in each of the nine graphs we ran *both* decisive SAT tests of Section 2:

- forced-different is decided by *merging* $s$ and $t$ (identifying the two vertices) and testing whether the merged graph is 5-UNSAT;
- forced-same is decided by *adding* the edge $st$ and testing whether the augmented graph is 5-UNSAT;
- if neither test returns UNSAT, the pair is free.

A per-solve conflict budget was used and tuned so that the number of indeterminate solves was driven to exactly zero; every pair received a decisive classification.

**Result.** All 1,955,948 non-adjacent pairs across the nine graphs were classified, and **every single one is free**: zero forced-same, zero forced-different, zero indeterminate. Table 1 reports the per-graph counts.

Table 1: Exhaustive forcing census over the realizable $\chi = 5$ lineage. For each graph, every non-adjacent pair was classified by two decisive SAT tests (merge for forced-different, add-edge for forced-same). Every pair is free.

| Order $n$ | Non-adjacent pairs | Forced-same | Forced-diff. | Free |
|---|---|---|---|---|
| 510 | 127,291 | 0 | 0 | all |
| 517 | 130,807 | 0 | 0 | all |
| 529 | 136,986 | 0 | 0 | all |
| 553 | 149,906 | 0 | 0 | all |
| 610 | 182,745 | 0 | 0 | all |
| 633 | 196,862 | 0 | 0 | all |
| 803 | 317,859 | 0 | 0 | all |
| 826 | 336,452 | 0 | 0 | all |
| 874 | 377,040 | 0 | 0 | all |
| **Total** | **1,955,948** | **0** | **0** | **all** |

Equivalently: in the realizable lineage, every non-adjacent pair admits both a same-colored and a differently-colored proper 5-coloring. All color forcing is *adjacency-local.* A single non-free pair would have been a candidate $\chi \geq 6$ ingredient (a forced-different pair is a clamp directly; a forced-same pair yields one by a splice). None exists.

## 5 The criticality explanation

The census of Section 4 is not a coincidence, and it is not even a separate fact. A criticality scan over the same nine graphs (deciding 4-colorability of $G - v$ for every vertex $v$, totaling on the order of 5,000 solves) found that *all nine lineage graphs are vertex-critical*: deleting any single vertex yields a 4-colorable graph, with zero redundant vertices and zero indeterminate verdicts.

**Theorem 1.** *Every graph in the realizable $\chi = 5$ lineage is vertex-critical, and therefore (by Corollary 1) contains no forced non-adjacent pair. Consequently the exhaustive census of Section 4 is a one-line corollary of the Essential-Pair Lemma.*

*Proof.* Vertex-criticality is the computational output of the scan. Given it, Corollary 1 applies to each of the nine graphs and yields that no non-adjacent pair is forced, which is exactly the census conclusion. □

**The conceptual heart.** Theorem 1 explains the census mechanistically, and the explanation is the central point of this note. The realizable lineage was *produced by SAT-driven minimization*: starting from a large $\chi = 5$ UDG, one repeatedly deletes vertices and edges while preserving $\chi = 5$, driving the graph toward a minimal core. Minimization of this kind drives a graph toward vertex-criticality. But by the Essential-Pair Lemma, criticality deletes *exactly* the vertices that could host a forced pair: a forced pair $(s, t)$ certifies that $s$ and $t$ are chromatically *redundant* (their deletion keeps $\chi \geq 5$), and a vertex-critical graph has no redundant vertices. The production process therefore removes every clamp-capable endpoint before any forcing search is ever run.

> *The realizable lineage is forcing-sterile by construction.* The missing $\chi \geq 6$ ingredient cannot be hiding in it, not because no one searched carefully, but because the very procedure that produced the lineage removed the only vertices that could carry the ingredient.

This has a concrete methodological consequence. A clamp (or any forced pair) can only live at chromatically redundant vertices, so a candidate-generation procedure aimed at $\chi \geq 6$ must *preserve* redundancy: it must **not** minimize toward criticality. Equivalently, any forcing sweep can be pre-filtered by an $O(n)$ criticality scan (keep only redundant vertices) rather than testing all $O(n^2)$ pairs, since forcing can only occur at redundant endpoints. We note one caveat carried from prior work: redundancy is necessary but not sufficient. There exist non-minimal 5-chromatic graphs (for instance a particular $C_6$-closure on 1155 vertices) that are forcing-free despite having redundant vertices.

## 6 A codegree obstruction to small dense hosts

If the missing $\chi \geq 6$ object is not in the realizable lineage (Sections 4-5), the alternative is to manufacture an abstract $\chi = 6$ graph and realize it as a UDG. The natural supply is $K_4$-free 6-chromatic graphs (one wants $K_4$-freeness because unit distances make $K_4$ geometrically rigid and hard to realize freely). These exist in abundance and can be made small. The obstruction recorded here is that they are *too dense* to be planar UDGs.

**The codegree bound.** In the Euclidean plane, two distinct unit circles meet in at most 2 points. Hence any two vertices of a planar UDG have at most 2 common neighbors: **every planar UDG is $K_{2,3}$-free**. (This is the classical fact underlying the Spencer-Szemerédi-Trotter $O(n^{4/3})$ bound on the number of unit distances among $n$ points [7].)

The $K_{2,3}$-free condition bounds the edge count. Counting paths of length two through each vertex,

$$\sum_v \binom{d_v}{2} \;=\; \#\{\text{paths of length 2}\} \;\leq\; 2\binom{n}{2},$$

because each unordered pair of vertices is the pair of endpoints of at most 2 such paths ($K_{2,3}$-freeness caps the common-neighbor count at 2). By convexity, $\sum_v \binom{d_v}{2} \geq n\binom{2m/n}{2}$, and solving the resulting quadratic in $m$ gives

$$m \;\leq\; \frac{n\big(1+\sqrt{8n-7}\big)}{4}. \tag{1}$$

We call the right-hand side the *codegree ceiling.*

**Manufactured hosts overshoot it.** Empirically, the smallest reachable $K_4$-free 6-chromatic (Folkman-type) hosts are dense, with edge-to-vertex ratio $m/n \approx 4.2$, far above the ceiling of (1). Concretely:

- an 18-vertex $K_4$-free 6-critical host has $m = 76$, against a ceiling of $\lfloor 57.17 \rfloor = 57$;
- the classical Mycielskian $M^3(C_5)$ on 47 vertices has 434 vertex pairs of common-neighbor count $\geq 3$, i.e. 434 violations of the $K_{2,3}$-free condition, with maximum codegree 11.

Such graphs cannot be planar UDGs; the violation is structural (it persists for edge-critical cores, which have no slack to remove), not an artifact of any particular construction heuristic.

**The violations are intrinsic to the critical core.** One might hope that the $K_{2,3}$ violations of a manufactured host are removable "padding" that a careful sparsification could shed while preserving $\chi = 6$. A top-down probe indicates otherwise. Starting from $M^3(C_5)$ (the 47-vertex $\chi = 6$ Mycielskian above, with 434 violating pairs) and repeatedly deleting edges while holding $\chi = 6$ as a hard constraint—re-establishing 6-chromaticity by a compensating forced-same edge whenever a deletion would drop it—drives the graph down to $m = 131$ edges, essentially the Kostochka-Yancey 6-critical floor $2.8 \cdot 47 - 1.8 = 129.8$. Yet the codegree excess does not fall to zero: the resulting edge-critical, $K_4$-free, $\chi = 6$ graph still has 166 vertex pairs of common-neighbor count $\geq 3$. Sparsifying all the way to the critical floor removes none of the $K_{2,3}$ violations; they live entirely in the 6-critical core. For this construction, then, the chromatic obstruction and the codegree obstruction are the *same* substructure: one cannot shed the codegree violations without destroying the very 6-chromaticity they accompany. (This is a heuristic observation about the $M^3(C_5)$-derived family, obtained by local search, not a theorem about all 6-critical graphs.)

**A rigorous core-size floor.** The $K_4$-free assumption can be dropped for a clean lower bound on the order of the 6-critical core. The Kostochka-Yancey theorem [3] states that every 6-critical graph has

$$m \;\geq\; 2.8\,n - 1.8.$$

Comparing this floor with the codegree ceiling (1), the floor *exceeds* the ceiling for $n \leq 12$ (and they cross at $n = 13$). Hence:

**Proposition 1.** *Any* 6*-chromatic unit-distance graph in the plane contains a* 6*-critical core on at least* 13 *vertices.*

*Proof.* A 6-chromatic graph contains a 6-critical subgraph $H$ (delete vertices and edges while $\chi = 6$). As a subgraph of a planar UDG, $H$ is $K_{2,3}$-free, so $|E(H)| \leq |V(H)|\big(1 + \sqrt{8|V(H)| - 7}\big)/4$. As a 6-critical graph, $|E(H)| \geq 2.8\,|V(H)| - 1.8$. For $|V(H)| \leq 12$ the lower bound exceeds the upper bound, a contradiction, so $|V(H)| \geq 13$. ☐

**The Folkman floor and the empty window.** The minimum order of a $K_4$-free 6-chromatic graph is the vertex Folkman number $F_v(2, 2, 2, 2, 2; 4)$. The literature settles this at

$$F_v(2, 2, 2, 2, 2; 4) = 16$$

(see Xu-Liang-Radziszowski [9]). Its only 16-vertex witnesses are the two Ramsey $(4, 4)$-graphs (Paley-type), which have $m \approx 60$. Both violate the codegree ceiling, since (1) gives exactly 48 at $n = 16$ and $60 > 48$. Therefore the UDG-necessary class, that is, the class of graphs that are simultaneously $K_4$-free *and* $K_{2,3}$-free, has no member at the Folkman floor. Because the ceiling (1) only reaches average degree 7.5 (the density of the Ramsey $(4, 4)$-graphs) around $n \approx 26$, even Folkman-floor density excludes the UDG-necessary class below roughly $n \approx 26$.

**Novelty.** We stress that the ingredients here are not new. The $K_{2,3}$-free fact is classical (it is the combinatorial heart of the unit-distance incidence bound [7]), the Kostochka-Yancey bound is a published theorem, and the Folkman number is a literature value. The contribution of this section is the *synthesis*: pinning the missing $\chi \geq 6$ object simultaneously below by criticality (Kostochka-Yancey), above by planarity-of-incidence ($K_{2,3}$-freeness), and at the abstract floor by Folkman, to conclude that the object cannot be a small dense host.

## 7 Discussion and the target object

The two preceding lines of argument pincer the missing ingredient from opposite sides.

- It cannot live in the realizable lineage (Sections 4-5): the lineage is vertex-critical and hence forcing-sterile, by the Essential-Pair Lemma.
- It cannot be a small dense Folkman-type host (Section 6): such hosts violate the codegree bound that every planar UDG must satisfy.

What remains is a single sharpened target.

> **Target object.** A 6-chromatic graph in the *UDG-necessary class* (simultaneously $K_4$-free and $K_{2,3}$-free), of order at least roughly 26, together with an explicit unit-distance embedding in $\mathbb{R}^2$.

In-class membership ($K_4$-free and $K_{2,3}$-free) is a *necessary* condition for UDG-realizability, not a sufficient one. A graph passing both local obstructions still requires an actual coordinate embedding to move $\chi(\mathbb{R}^2)$. The codegree ceiling, in particular, is a counting bound: the true maximum number of unit distances among $n$ points sits well below it at small $n$, so a graph that respects (1) need not be realizable.

**Honest limitations.** We list the caveats explicitly.

1. *The Essential-Pair Lemma is elementary and known.* Its corollary is Theorem 3.17 of Martin [4] (phrased there via "implicit-edges" and "implicit-identities"), and it is implicit in standard critical-graph theory [2]. We include a proof only for self-containedness and make no priority claim.
2. *The $K_{2,3}$-free fact is classical.* It is the basis of the Spencer-Szemerédi-Trotter unit-distance edge bound. The novelty of Section 6 is the synthesis with Kostochka-Yancey and the Folkman number, not the ingredients.
3. *The density observation is empirical.* The claim that reachable $K_4$-free 6-critical hosts sit at $m/n \approx 4.2$ is an observation over roughly 35 cores from one construction family, not a theorem about all $K_4$-free 6-critical graphs. Only the $n \leq 12$ exclusion (Proposition 1) and the Folkman-floor exclusion are rigorous.
4. *No bound on $\chi(\mathbb{R}^2)$ moved.* This note is negative and explanatory. It tightens the description of where the missing object can live; it does not produce the object.

**Outlook.** The results suggest a clear procedural lesson: generate *inside* the UDG-necessary class rather than above it. Manufacturing $K_4$-free 6-chromatic graphs and cutting them down fails the codegree bound; the alternative is to enforce $K_4$-freeness and $K_{2,3}$-freeness as hard invariants from the start, seeding from a real $\chi = 5$ UDG (which lies in the class automatically) and adding only codegree-safe and $K_4$-safe structure until the chromatic number is forced upward. Whether that class forces $\chi \geq 6$ at some attainable order, or instead caps at $\chi = 5$ (the UDG analogue of $\chi(\mathbb{Q}^2) = 2$), is the open question this framing isolates.

Preliminary computational probes of this question are suggestive but inconclusive, and they approach the target object from three independent directions, none of which reaches it. *(i) Growth from below.* A greedy search that grows a graph from the empty graph by adding only codegree-safe and $K_4$-safe edges caps at $\chi = 4$ across orders $17 \leq n \leq 64$: it never even reaches $\chi = 5$. *(ii) Annealing.* A local search with both edge additions and deletions (simulated annealing over graphs kept $K_4$-free and $K_{2,3}$-free, minimizing the number of proper 5-colorings) over orders $26 \leq n \leq 40$, with on the order of 20,000 restarts, reaches in-class $\chi = 5$ at orders as small as $n = 29$ but never $\chi = 6$, with the edge density plateauing near 84% of the codegree ceiling (1) at every $n$. *(iii) Repair from above.* The top-down probe of Section 6, which starts from a genuine $\chi = 6$ graph ($M^3(C_5)$) and repairs its codegree violations while holding $\chi = 6$, stalls with the violations intact in the 6-critical core. All three are heuristic, not non-existence results: a $\chi \geq 6$ in-class graph may live at larger $n$, or at configurations none of the three local searches reach. But together they triangulate the same conclusion—the target object, if it exists in this range, is not easily constructed by local moves from any natural seed.

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